the engine runs at the idling speed. So that when the engine speed gets raised, pump speed also get raised and sends oil with high pressure. This causes more oil to get wasted if the oil seals and joints are damaged. To avoid this, pressure relief valve is fixed in this pump. This valve releases the excess pressure developed inside the pump. The following are the different types of pressure relief valves used in automobiles.

Based on design it is classified into two types they are

1. Ball type pressure relief valve
2. Plunger type pressure relief valve

8.6.1 Ball Type Pressure Relief Valve

In this type a ball valve, spring and adjustment screw are present. The ball valve will be placed in its seat supported with a spring. All these above components are placed inside the pump housing. The figure 8.6.1 shows the view of the ball type pressure relief valve.

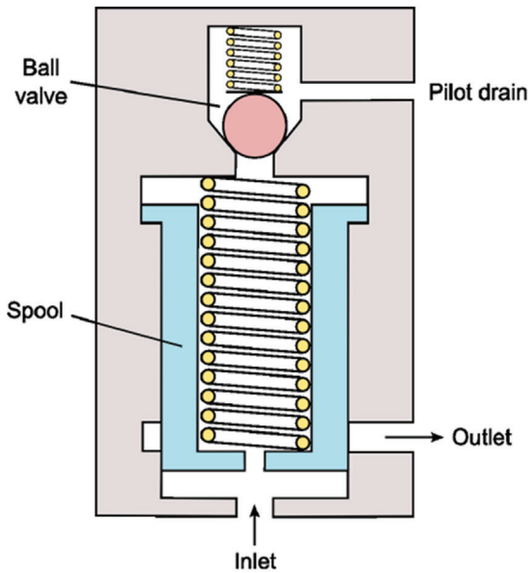


Figure 8.6.1 Ball Type Pressure Relief Valve

Working:

When the engine is off, the spring forces the ball to seat on its seat tightly. When the engine is started, based on the oil pressure and the spring tension the ball value will be in the opened or closed position. When the engine speed is increased, the pressure of oil is also increased. The increased pressure once if it reaches the fixed level, it presses the spring so that it moves the ball from the seat and releases the pressure and the excess oil reaches the inlet via bypass or to the oil sump.

Adjusting Screw And Lock Nut:

Adjusting screw is used to maintain the required level of pressure. If the screw is tuned inwards it increases the spring tension and allows more pressure to open the ball valve. If the screw is tuned outwards it reduces the spring tension and makes the pressure of oil to open the ball valve easily. Normally for the engines when runs at 1000 rpm at 43 degree Celsius the spring tension will be of 2.5 kg/cm^2.

8.6.2 Plunger Type Pressure Relief Valve

It is similar to the ball type valve but has a plunger, spring and adjustment shims. The plunger is placed on the seat with the help of the spring tension (Fig 8.6.2).

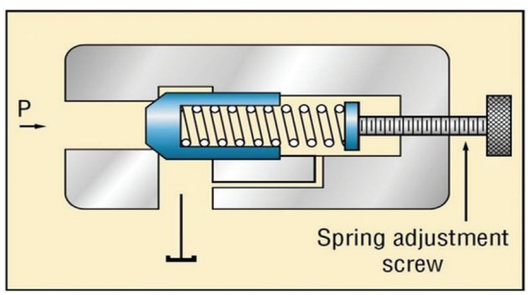


Figure 8.6.2 Plunger Type Pressure Relief Valve

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